11/09/2024

Key Biometric process

- Enrollment process
- Recognition process
  - verification (1:1 matching)
  - Identification (1:N matching)
  - Template Update & maintenance

Biometric Modality — physiological / Behavioural

characteristics

(1) Universality
(2) Uniqueness
(3) collectability
(4) performance
(5) Acceptability
(6) circumvention
(7) permanance

A biometric s/m consists of 4 main process:

(1) Enrollment
↳ Capturing & storing

(2) ~~Feature Extraction~~ steps
1. Data Capture
2. Quality check
3. Registration

(2) Feature extraction
↳ converting biometric data into unique digital template
1. preprocessing (ring enhancement, noise reduction, normalize)
2. Feature detection (minutiae points)
3. Template creation. (converting to mathematical representation)

(3) Template Storage.
↳ securely storing the biometric template
1. Encryption (Template is encrypted for security
2. Database security (stored in secure database (local or cloud)
3. Indexing (Template are indexed for fast retrieval)

(4) Matching process.
↳ comparing live biometric data with stored templ
1. Live Capture (New biometric sample is Captured)
2. Feature Extraction
3. comparision
4. Decision

Types of Matching — verificatn / Identificatn

verification (1:1)
↳ confirms claimed

13/08/

Key challenges

* Environmental factors (lighting conditions, noise, weather
* User cooperation (proper positioning, movement
* Security Threads (spoofing attacks, fake biometrics
* privacy concerns (Data protection, unauthorized access)

→ List down the different Biometric modality and their characteristics.

Biometric identifier	Universality	Uniqueness	collectability	performance	Acceptability	circumvention	performance
DNA	High	High	low	High	low	low	High
Ear	Medium	Medium	Medium	Medium	High	medium	H
Face	High	low	High	low	High	High	M
Facial Thermogram	High	High	High	medium	High	low	L
Fingerprint	medium	High	medium	high	medium	medium	H
Gait	M	L	H	L	H	M	L
Hand Geometry	M	M	H	M	M	M	M
Hand vein	M	M	M	M	M	M	M
Iris	H	H	M	H	L	L	H
keystroke	L	L	M	L	M	M	L
odor	H	H	L	L	M	L	H
palmprint	M	H	M	H	M	M	H
Retina	H	H	L	H	L	L	M
Signature	L	L	H	L	H	H	L
voice	M	L	M	L	H	H	L

18/08/2016 Performance measures in biometric systems.

FAR
False Acceptance Rate
↳ security measure

FRR
False rejection Rate
↳ Usability measure

FTE
Failure to enroll
↳ System Universality

EER
Equal error rate
↳ Overall performance

ATV
Acquisition Time to verification
↳ system speed.

Performance

FAR (FMR ~~FMR~~ – False match Rate)

* probability that the system incorrectly accepts an unauthorized user.

$$FAR = \frac{\text{No. of false Accepts}}{\text{Total no. of imposter Attempts}} \times 100$$

Eg: Fingerprint s/m incorrectly accepts 5 unauthorized user out of 100;000 imposter

$$= \frac{5}{100,000} \times 100 = 0.005$$

FRR (FNMR – false non-match Rate)

* probability at which the s/m incorrectly reject an authorized users.

$$FRR = \frac{\text{No. of false Rejects}}{\text{Total no. of Genuine Attempts}} \times 100$$

Range : 1% – 5%.

FTE

* percentage of users who cannot successfully enroll in the biometric s/m.

$$FTE = \frac{\text{No. of failed enrollments}}{\text{Total enrollment attempts}} \times 100.$$

• High FTE reduces s/m universality.

EER

* The point where false acceptance Rate equals false reject rate.
* Also called Crossover Error Rate
* Single metric for s/m comparison.
* Lower EER = Better performance

(+) ~~plot FAR & FRR curves against~~

ATV

* Total time from biometric presentation to verification decisio

ATV = Capture Time + processing Time + Decision Time + Response time

Factors Affecting:

- Sensor technology and quality
- processing power & algorithm
- N/w latency
- Database size

performance Trade Offs

Fundamental challenge:

* Security vs. Usability Trade off: FAR and FRR are inversely related.

High Security (Low FAR)

* Stricter matching thresholds
* High FRR (more rejections)
* Better for high-security application
* May frustrate legitimate users

High Usability (Low FRR)

* Relaxed matching thresholds
* High FAR (more false accepts)
* Better user experience
* potential security vulnerabilities

Applications of Biometric Systems

- Access control
  - Building access, computer login
- Government
- Financial service
- Time & attendance
- Healthcare
- Mobile Consumer
- Law enforcement

Access control & physical security

Applications

- Building access control
- computer & N/w login
- safe & vault access
- vehicle access s/m
- Server room security.

Requirements

- Low FAR for security
- Reasonable FRR for convenience.
- Fast verification (low ATV)
- Durability & reliability
- Integration capabilities

Common Technologies

Fingerprint, Iris. etc.

Security Vulnerabilities

(1) Spoofing Attacks

↳ Attempts to fool biometric sensors using fake traits

* Fingerprint : Silicon molds, printed patterns.
* Face : Photos, videos, 3D masks
* Iris : printed patterns, contact lenses.
* Voice : Recorded audio, synthesis

(2) Template Security

- Template theft from database
- Template modification
- Cross matching b/w stns.
- Biometric reconstruction from templates.

(3) Transmission Security
- Man-in-the-middle attack
- Replay attacks with intercepted data.
- Protocol vulnerabilities.

Security Countermeasures

Anti-spoofing Measures — Template protection

- Liveness Detection: pulse, blood flow

Hybrids

Physiological vs Behavioral

→ Physiological
- Fingerprint (Ridges & valleys)
- Face geometry
- Iris Texture
-

→ Behavioral
- signature dynamics
- Gait (walking pattern)
- key Dynamics
- voice pattern

(Non-contact: face, Iris & voice) (Fingers contact)

Fingerprint Fundamentals

It is a pattern of ridges and valleys on the fingertip surface, formed permanently during fetal development.

① Unique
- No 2 individuals have same ridge patterns.

② Permanent
- How long it last

③ Universal.
- present in every

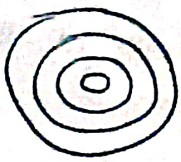
Global pattern Classifications

① Loop



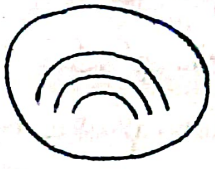
60-65%.
- Ridges enters & exit at same side

② Whorl



30-35%.
- Ridges forms a circular

③ Arch

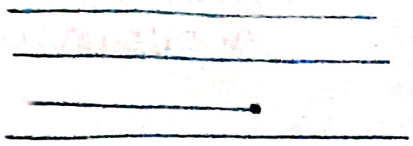


5%.

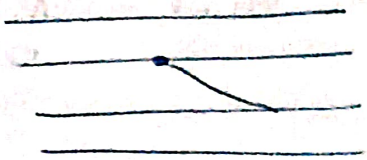
Minutiae : The Basics of Matching

They are the local ridge discontinuities. A typical print contains 30–70 minutiae; matching algorithms usually need 8–20 correspondence to declare a match.

Ridge ending



Ridge Bifurcation



Finger-Scan S/M pipeline

1. Image Acquisition
2. preprocessing – Thinnings / Segmentation / Enhancing / Binarization
3. Feature Extraction – Detect & extract the minutiae points
4. Template Generation –
5. Matching Engine – Compares live vs stored templat
6. Decision – Accept / Reject

Fingerprint Sensor Technologies at a Glance

1. optical (FTR) → LED light + prism → Door access
   ridges scatter light & valleys reflect internally.
2. Capacitive → Micro-capacitors array → Smartphones.
   measures capacitance difference b/w ridge Contact & valley gap
8. Ultrasonic → Sound pulses reflect off ridges/valleys and sub-surface tissues → under display.

Optical Vs Capacitive Sensing

- Optical (FTIR)
- Capacitive (Solid State)

Ultrasonic Vs Multispectral Sensing

Matching Algorithms Approaches

Approach	Principle	Notes
Minutiae based	→ Aligns & compares set of minutiae via. point pattern matching	→ Industry standard - ANSI/NIST, ISO/IEC 19794-2
correlation based	→ Overlays two images directly; Computes pixel wise correlation	→ sensitive to translation / rotation / distortion; less common now.
Ridge feature (Texture)	→ Gabor filter bank produces a texture feature vector ("Finger code")	→ useful when minutiae extraction is unreliable.
Deep learning Based	→ CNN / siamese n/w embeddings learned directly from ridge images.	→ Emerging - robust to distortion, partial & cross sensor prints.

Performance Matrics.

FAR	FRR	EER	FTE	~~FTA~~	FTA
probability at which imposter is strongly accepted.	probability at which genuine user is strongly rejected.	point at which FAR=FRR lower is better	% of users whose prints can't be capture well		Probability to capture a use like service.

Fingerprint Vs Other physiological Modalities

Strength & weakness of finger-scans.

Strengths

- Mature, Standard (ISO/IEC 19794-2)
- low cost per sensor - Enable mass deployment

weakness

- Non-universality - some users produce unreadable prints
- Spoofing risks on sensors lacking liveness detection

1/9/26 Facial Recognition

(1) Geometry Based Approach (Feature based approach)
- Measuring the facial landmarks.
- Distance from eyebrows or

(2) Holistic Approach or Eigen faces or fisher faces. (LDA method)
- *LDA approach → linear discriminant Analysis
- (*) LBP - local Binary pattern
  - encode local ~~binary~~ micro texture around the face
- (*) Deep CNN embeddings
  Matric learning loses ~~bases~~ transmit matric fales to embedding
- (5) 3D or Depth cells.
  - Using antispoofing
  - projecting infrared dot patterns to get depth

Strengths

* contactless & remote
* works with widely available RGB camera.
* High user acceptance
  - feeds natural as well as non invasive
* Milliseconds speed CNN embedding matching
* Handles 1:1 verification and 1:N identification

weakness

* ~~easy to spoof~~ sensitive to pose, eliminated, expressions, occlusion
* Aging degrades accuracy without reenrollment.
* easy to spoofs by photo, video, Deepfake without ~~live~~ liveness

Iris

Basic features of Iris

- Texture is the unique feature
- Mesh work, crypt work, Furrows, Freckles, collarette there are the terms which are formed at the fetal stage and they are unique even to the identical twins.

~~IRIS~~

- NIR cameras, visible cameras are been used. (bw 700-900 nms) [Even in dark light eye will get image]

Segmentation

- Daugment's method is widely used. (Daugment operator)
- Interdio differential operator
- Rubber sheet model.

Matching → Hamming distance algorithm

Strength	weak
* Extremly high accuracy	* Expensive
* EER is low below 0.1%.	* User operation
* Stable	* Contactless.
* very large feature space	* Degrading by catarat
* contactless, hijgienic consent	
* Hard to surgical alter; without vision machine damage	

physiological characteristics	Behavioral factors
• Vocal track	• Speaking rate face
• Larynx	
• Nasal cavity shape	

Image acquisition
• Capture through microphone
preprocessing
• Noise reduction, voice activity detection, pre-emphasis
Feature extraction
• MFCC - Mel frequency cepstral coefficient
↳ take dominant voice. (with this help feature is extracted)

Deep Neural Network (DNN)

Matching
• ~~Score~~ scoring
↳ cosine similarity

. Text dependent phrase as well text independent
• Text prompted.
↳ Anti replay attacking s/m.

• GMM (combine universal background model.)
• UBM (Universal background model)

Eigen vector + PLV

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Retina Vs palm scan (Hand Scan)

• Hand Geometry.
(1) Lengths & widths of fingers
(2) Height and widths of palm
(3) Distance b/w finger joints
(4) Curvature of fingers
(5) Overall hand thickness

Key components:

• platen
↳ alignment, pleg position at hand consistently the camera captures the top down ~~clone~~.

• ~~Mirror~~ camera
• Mirrors → ~~#~~Angled mirror
↳ ~~Encrypted~~ image is provided.

- Infrared
  ↳ capture the depths

* DSP or micro controller (ing in appropriate way)
  ↳ extract the actual measurement

S/m Pipeline

(1) Enrollment
Feature extraction
template generation

→ where it is being used ;

(1) Nuclear power plants & secure sites
(2) Some universities
(3) Disney Theme park

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Strengths

- Fast process in 1 sec
- Template size is very less
- Unaffected by dirt
- works well in outdoor, industrial areas.

weakness

- low uniqueness. 1:N identifications
- Hand deformity, change by age.
- Bulky.
- Can't be taken for forensic identification

Retina

- Blood vessels of retina is taken and that is the unique part considered in the retinal scan
- live sensitive tissue at the back of eye which is covered by a capillary pattern formed randomly during fetal development itself

Key components

① IR cameras : to capture the image and IR lights.

② optical allignment of eye piece.

③ photo sensitive sensor - captures the reflected light around the circular

④ Scanning Mechanism - Traces all the 360° patterns.

⑤ Image processor

S/No pipeline

① Enrollment (position) - by clearly placing ie., the position

② allignment and focus - Only by focussing we will get clear image. place it in a fixed place.

③ Illuminate - low infrared beam is projected because (it will affect the eye)

④ Scan - the optical nerve part, after setting all other parts still then scan.

Feature extraction
template
matching
decision

* Eye dentify is the earliest commercial available retinal scanner in banking (Eye dentify KAM 2001)

Strengths
- Extremely low FAR
- unique
- Stable
- cannot be spoofed easily
- may high resistant to colour

weakness
- Bulky enviroment.
- Highly intensive
- Expensive
- Enrollment is the slow process
- unintended health information
- privacy consent

	Hand Scan	Retinal Scan
Uniqueness	Low	High
accuracy	Moderate	High
Speed	Fast (1 sec)	Slower (4 to 5 sec)
User comfort	High	low
Hardware cost	low - Moderate	High
Stability over time	Changes	Stable
Spoofing resistance	low - Moderate	High
Typical security time	Low - Moderate	High

AFIS

Automated fingerprint identification